SE lab 7: Fuzzing for Input Validation Bugs

Name: John S Thomas
SRN: PES2UG24CS207

Step 2: Run the buggy application (python food_delivery.py)

```
(.venv) bane@BanePro Lab 7 % cp food_delivery_buggy.py food_delivery.py
python food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:2,burger:1
Order: {'pizza': 2, 'burger': 1}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 4.5
ETA in Minutes: 33

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE50
Discount: 50
(.venv) bane@BanePro Lab 7 %
```

Step 4: Run test_food_delivery.py using Hypothesis (pytest failing tests)

```
(.venv) bane@BanePro Lab 7 % pytest test_food_delivery.py
============================= test session starts ==============================
platform darwin -- Python 3.15.1, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/bane/Downloads/L6_L7_20_food_delivery/Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_food_delivery.py FFF                                                [100%]

=================================== FAILURES ===================================
_____________________________ test_parse_order_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/bane/Downloads/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 5, in test_parse_order_no_crash
  |     def test_parse_order_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bane/Downloads/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 7, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~^^^
    |   File "/Users/bane/Downloads/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 5, in parse_order
    |     order[name.strip()] = int(qty)
    |                           ~~~^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/bane/Downloads/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 9, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with: {e}"
    | AssertionError: parse_order crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/bane/Downloads/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 7, in test_parse_order_no_crash
```

Step 5: Run the fixed application (python food_delivery.py)

```
(.venv) bane@BanePro Lab 7 % cp food_delivery_fixed.py food_delivery.py
python food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:2,burger:1
Order: {'pizza': 2, 'burger': 1}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 4.5
ETA in Minutes: 33

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE50
Discount: 50
(.venv) bane@BanePro Lab 7 %
```

Step 6: Re-run test_food_delivery.py on fixed code (pytest passing tests)

```
(.venv) bane@BanePro Lab 7 % pytest test_food_delivery.py
============================= test session starts ==============================
platform darwin -- Python 3.15.1, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/bane/Downloads/L6_L7_20_food_delivery/Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_food_delivery.py ...                                                [100%]

============================== 3 passed in 0.20s ===============================
(.venv) bane@BanePro Lab 7 %
```